

Model Performance Summary

A Decision Tree/Logistic Regression model was developed to predict employee attrition. The model achieved an overall accuracy of 87% on the test dataset.

The model performed exceptionally well in identifying employees who are likely to stay with the organization, achieving a recall score of 96% for the non-attrition class. However, the model showed moderate performance in identifying employees likely to resign, with a recall score of 38%.

The precision score for attrition prediction was 67%, indicating that when the model predicts an employee is likely to leave, it is correct approximately two-thirds of the time.

While the overall accuracy is satisfactory, the low recall for attrition cases suggests that additional model tuning, class balancing techniques, or ensemble methods may be required to improve the detection of employees at risk of leaving.

Overall Accuracy: 87%

Attrition Precision: 67%

Attrition Recall: 38%

Attrition F1-Score: 49%